Point-Source versus Body-Force Galerkin Representations of an Elastic Cubic Scatterer: a Mie- and Kennett-Validated Hierarchy

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Abstract

We ask what a body-force Galerkin representation of an elastic cubic scatterer buys over a simple point source, and answer it from reproducible evidence rather than assumption. Three single-site representations—the Born point, the Eshelby-corrected point with a radiation-reaction local density of states (LDOS), and a nine-component body-force Galerkin closure (T_9)—together with a point versus volume-averaged inter-site propagator, are placed in a hierarchy validated against two exact oracles: the elastic Mie sphere (single-site far field, partial-wave reconstructed effective contrasts) and the Kennett layered reflectivity (periodic-slab R_{PP}). On a controlled cost–accuracy sweep against the equal-volume Mie sphere (three contrasts × four ka values, P-incidence) the relative far-field error orders as eshelby < t9 < born in all twelve cells, while the wall-time orders as born < eshelby < t9 with T_9 roughly 10× costlier than the Eshelby point: the Galerkin closure is dominated on both cost and accuracy. This reflects a cube-versus-sphere geometric mismatch faithfully imprinted by the finite-size cube form factor (itself Kennett-validated to ∼1 %), not a defect in the Galerkin model. The evidence-derived verdict is that a point/Eshelby representation suffices single-site, that volume averaging and the quadratic (T_{27}) extension are not repaid inter-site ($\leq 0.07\,\%$), and that the genuine value of the body-force Galerkin formulation lies in its singularity-free weak form and in the Mie-validated radiation-reaction attenuation it supplies—in the self-energy, not in the observable far field.

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1 Introduction

Elastic wave scattering from discrete heterogeneities—cracks, inclusions, cavities, mineral grains—underlies seismic attenuation, coda formation, and the effective elastic properties of composite and granular media. The natural framework is the elastodynamic volume integral equation (VIE), or Lippmann–Schwinger equation, in which the scattered field is generated by an equivalent source distribution supported on the heterogeneity: a body-force term proportional to the density contrast and a stress term proportional to the stiffness contrast [1, 2]. Collecting the response of a single inclusion into a T-matrix T_0 and coupling many inclusions through the background Green's tensor yields the Foldy–Lax multiple-scattering system [3, 4, 5], the elastodynamic analogue of the coupled-dipole method familiar from electromagnetics [6].

A modelling choice sits at the heart of this construction: *how should the single inclusion be represented?* At one extreme the inclusion is a *point source*—a force monopole and stress dipole carrying the volume-integrated contrast, optionally dressed by an Eshelby concentration factor [7] and a radiation-reaction term. At the other extreme it is resolved as a finite body through a *Galerkin projection* of the VIE onto polynomial trial functions over the inclusion volume, which captures the spatial variation of the internal field and removes the singular kernels of the point-evaluation route. The point representation is cheap and shape-agnostic; the Galerkin representation is more complete but carries finite-size form factors, internal amplification, and a heavier per-site cost. For an elastic *cubic* scatterer—the geometry of interest for cracked and blocky media—this paper asks a sharp question: **does the added complexity of a body-force Galerkin closure buy observable far-field accuracy over a point source, and where (if anywhere) is its value real?**

We answer this by building a hierarchy of representations and validating each rung against exact references—the elastic Mie sphere for the single-site far field and effective contrasts, and the Kennett layered reflectivity for the periodic-slab reflection coefficient—then reading the verdict off a reproducible cost–accuracy sweep. The hierarchy and its validation are the substance of the paper; the verdict is its conclusion.

Contributions. We separate standard ingredients from the novel results of this work.

- *Standard.* The Lippmann–Schwinger VIE and its body-force/stress decomposition [1, 2]; the static Eshelby concentration tensor [7]; the Foldy–Lax multiple-scattering system [3, 4, 5]; the elastic Mie partial-wave solution [8]; the Kennett layered reflectivity recursion [9]; and the Aki–Richards moment-tensor radiated-power formula [10].

- *Novel (this work).* (i) The cube $-1/3$ finite-size form-factor correction to the effective modulus contrast, validated against Kennett to a $\sim$1 % floor (Section 4.1); (ii) the strain-channel radiation reaction identified as a strain LDOS, $\mathrm{Im}[\Delta c^*]$, derived in closed form and validated against the exact Mie a_0/a_2 coefficients (Section 4.3); (iii) the corrected density radiation reaction $\mathrm{Im}[\Gamma_0]$ with the full isotropic $(1/3)P + (2/3)S$ LDOS weights, replacing an earlier overdamped pure-S form (Section 4.3); and (iv) the *measured* verdicts—that the Eshelby point beats the T_9 Galerkin closure on single-site far-field cost–accuracy versus an equal-volume sphere (Section 6), and that the inter-site quadratic coupling contributes $\leq 0.07\,\%$ (Section 5).

The quadratic, 27-component Galerkin extension is scoped out of this paper: it furnishes no Mie-consistent single-site far-field gain and a negligible inter-site contribution, so it appears only as the excluded top rung of the hierarchy (Sections 2.3 and 5.3). The remainder of the paper develops the background theory and oracles (Section 2), the T_9 Galerkin closure (Section 3), the single-site self-energy corrections (Section 4), the inter-site coupling analysis (Section 5), the cost–accuracy synthesis (Section 6), and the evidence-derived conclusion (Section 7).

2 Background theory

2.1 Volume integral equation and the Lippmann–Schwinger formulation

Consider an isotropic, homogeneous elastic background with Lamé parameters (λ_0, μ_0), density ρ_0, and corresponding Green's tensor $G_{ij}(\boldsymbol{r})$ (the full elastodynamic propagator at angular frequency ω). A finite inclusion $V \subset \mathbb{R}^3$ introduces contrasts Δc_{nklj} in the stiffness tensor and $\Delta\rho$ in the density relative to the background. The total displacement $u_i(\boldsymbol{r})$ satisfies the Lippmann–Schwinger (LS) volume integral equation [1]:

$$u_i(\boldsymbol{r}) \;=\; u_i^0(\boldsymbol{r}) + \omega^2 \int_V G_{ij}(\boldsymbol{r}-\boldsymbol{r}')\,\Delta\rho\, u_j(\boldsymbol{r}')\,dV' - \int_V \partial'_k G_{in}(\boldsymbol{r}-\boldsymbol{r}')\,\Delta c_{nklj}\,\partial'_l u_j(\boldsymbol{r}')\,dV', \quad (1)$$

where u_i^0 is the incident displacement and the primed derivative ∂'_k acts on the source variable $\boldsymbol{r}'$. The first integral represents the body-force coupling through density contrast; the second represents the elastic stiffness contrast through a double-gradient of the Green's tensor.

The elastodynamic Green's tensor has the near-field singularity structure

$$G_{ij}(\boldsymbol{s}) \sim |\boldsymbol{s}|^{-1}, \qquad \partial G \sim |\boldsymbol{s}|^{-2}, \qquad \partial\partial G \sim |\boldsymbol{s}|^{-3}, \quad (2)$$

so the body-force kernel G_{ij} is locally integrable, but taking field-point derivatives of the stiffness-contrast term introduces increasingly singular kernels. Evaluating the stiffness integral at the field point inside V requires either a Cauchy principal-value regularization or, for quadratic and higher trial functions, Eshelby-type delta-function corrections [7]. This singular structure is the central technical constraint that shapes the hierarchy of approximations discussed in Section 3.

The far-field amplitude radiated by an inclusion is compactly encoded in the single-site T-matrix T_0, which maps the incident wave onto the scattered field:

$$u_i^{\mathrm{sc}}(\boldsymbol{r}) \;\sim\; G_{ij}(\boldsymbol{r}-\boldsymbol{r}_0)\,T_{0,jk}\,u_k^{\mathrm{inc}}(\boldsymbol{r}_0) \qquad (|\boldsymbol{r}-\boldsymbol{r}_0| \to \infty). \quad (3)$$

When multiple inclusions are present, the coupled system is governed by the Foldy–Lax equations [3, 4], in which T_0 feeds the inter-site propagator $G_0(\boldsymbol{r}_{ij})$ (Figure 2).

2.2 Exact reference solutions

All representations examined in this paper are validated against two exact reference solutions, treated as oracles.

Elastic Mie sphere. For a spherical inclusion of radius a and contrast $(\Delta\lambda, \Delta\mu, \Delta\rho)$, the scattered displacement admits an exact partial-wave (Mie) expansion [8]. An incident P-wave along $\hat{\boldsymbol{z}}$ is decomposed into Legendre orders n; at each order the Helmholtz decomposition yields a 4×4 boundary system (displacement and traction continuity for P- and SV-channels) whose Cramer-rule solution gives the scattered-field amplitudes a_n (P-to-P) and b_n (P-to-S). At finite contrast the series converges for all ka; no linearization is invoked.

Effective contrasts are extracted from the partial-wave coefficients by Legendre projection: the monopole a_0 yields the bulk-modulus contrast, the dipole a_1 yields the density contrast, and the quadrupole a_2 yields the shear-modulus contrast. These Mie-reconstructed values serve as the ground-truth targets against which point and Galerkin representations are compared in Sections 4–6.

Kennett layered reflectivity. For the periodic-slab geometry, the exact specular P-to-P reflection coefficient R_{PP} is computed via the Kennett recursion [9], which assembles the full reflection matrix of a stratified medium using propagator matrices and a generalized Bremmer series. At normal incidence with a single embedded layer of thickness H and contrast $(\Delta\lambda, \Delta\mu, \Delta\rho)$, the Kennett formula provides a frequency-dependent $R_{PP}(\omega)$ valid at all contrasts. This serves as the slab oracle in Section 6.

Both references are free of any T-matrix approximation and are therefore well-suited as error baselines: any discrepancy between a representation and an oracle is a measure of that representation's truncation error.

2.3 The single-site hierarchy

The paper compares three single-site representations of a cubic scatterer, ordered below by increasing model complexity and computational cost (Figure 1). Whether this added complexity reduces the observable far-field error is the central question, answered in Section 6.

1. **Born / point-source.** The simplest approximation replaces the internal field by the incident field, $u_j(\boldsymbol{r}') \approx u_j^0(\boldsymbol{r}')$, and integrates directly. This gives a strength proportional to $V\Delta c$ or $V\omega^2\Delta\rho$, with no self-interaction. The result is the leading-order Born scattering amplitude: a point source with volume-weighted contrast but no radiation-reaction damping.

2. **Eshelby-corrected point scatterer.** The Born strength is renormalized by the on-site radiation reaction $\operatorname{Im}\Gamma_0$, where Γ_0 is the imaginary part of the volume-averaged background Green's tensor evaluated at zero separation. This correction, which follows from the elastodynamic optical theorem, accounts for the energy radiated by the scatterer and is essential for the point representation to satisfy energy conservation at finite frequency. The real part of Γ_0 encodes the static Eshelby concentration tensor [7], which dresses the incident strain inside the inclusion.

3. **Body-force Galerkin T_9.** The LS equation (1) is projected onto the nine-dimensional trial space spanned by the three uniform displacement modes and the six independent strain modes. Because the trial functions are polynomials of degree ≤ 1, the bilinear forms involve only the body-force integral $\iint G_{ij}\, dV\, dV'$ (kernel $|\boldsymbol{s}|^{-1}$, absolutely convergent) and the stiffness integral with one derivative $\iint \partial'_k G_{in}\, dV\, dV'$ (kernel $|\boldsymbol{s}|^{-2}$, conditionally convergent via integration by parts to the surface). No Eshelby delta correction is required:

the Galerkin projection absorbs the singularity into well-posed five-dimensional surface integrals. This is the lowest-order closure that correctly captures both displacement and strain amplification inside the cube, giving an $\mathcal{O}(ka)^2$ accurate T-matrix across the Rayleigh regime.

A natural further rung—the 27-component Galerkin extension with quadratic trial functions—is excluded here: it yields no Mie-consistent single-site far-field gain and contributes $\leq 0.07\,\%$ inter-voxel coupling (Section 5).

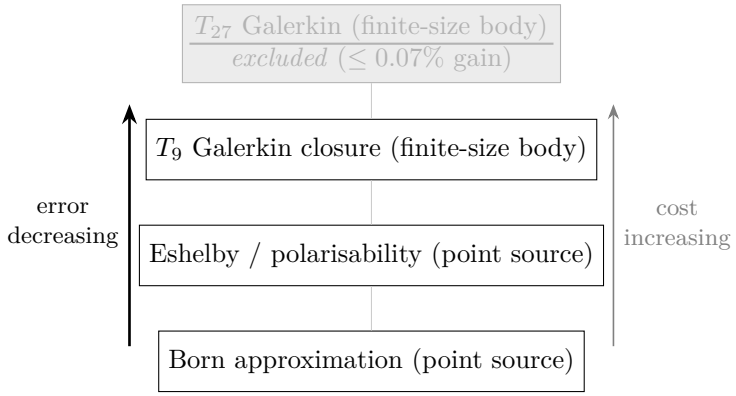


Figure 1: Ladder of approximations from lowest (Born, point source) to the finite-size Galerkin closure T_9. The T_{27} rung is shown greyed and struck out because measurements show it contributes $\leq 0.07\,\%$ gain over T_9 at all validated frequencies, making the added cost unjustified. Rungs are ordered by increasing model complexity and computational cost (Born $<$ Eshelby-point $< T_9$). Whether far-field accuracy follows this ordering is the question settled empirically in Section 6; it does not, for the single-site far field.

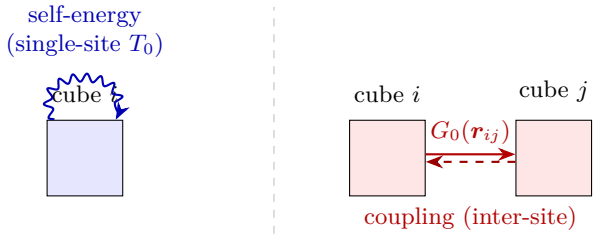


Figure 2: Single-site self-energy (left: scatterer i radiates back into itself, captured by T_0) versus inter-site coupling (right: wave field propagates from scatterer i to j via the background Green tensor $G_0(\boldsymbol{r}_{ij})$ and back). The Foldy-Lax system couples both contributions.

3 The body-force Galerkin closure (T_9)

3.1 Galerkin projection onto the polynomial trial space

Let $V = [-a, a]^3$ be the cubic inclusion. The *nine-dimensional trial space*

$$T_9 \;=\; \left\{ \boldsymbol{\varphi}(\boldsymbol{r}) \;\middle|\; \varphi_i(\boldsymbol{r}) = U_i + \varepsilon_{ai}\, r_a, \quad \varepsilon_{ai} = \varepsilon_{ia} \right\} \tag{4}$$

spans the three uniform-displacement modes (3 degrees of freedom from U_i) and the six independent components of the symmetric strain tensor (6 degrees of freedom from ε_{ai}), giving $3+6=9$ trial functions in total. The trial functions are polynomials of total degree ≤ 1 in $\boldsymbol{r}$.

Definition 1 (T_9 Galerkin closure)**.** The T_9 closure is the variational problem: find $\boldsymbol{u}^h \in T_9$ such that for all $\boldsymbol{\varphi} \in T_9$,

$$\langle \boldsymbol{\varphi}, \boldsymbol{u}^h \rangle_V \;=\; \langle \boldsymbol{\varphi}, \boldsymbol{u}^0 \rangle_V + \omega^2 \Delta\rho\, \mathcal{B}_{\mathrm{body}}[\boldsymbol{\varphi}, \boldsymbol{u}^h] - \Delta c_{nklj}\, \mathcal{B}_{\mathrm{el}}[\boldsymbol{\varphi}, \boldsymbol{u}^h], \tag{5}$$

where the bilinear forms are

$$\langle \boldsymbol{\varphi}, \boldsymbol{u} \rangle_V := \int_V \varphi_i(\boldsymbol{r})\, u_i(\boldsymbol{r})\, dV, \tag{6}$$

$$\mathcal{B}_{\mathrm{body}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, u_n(\boldsymbol{r}')\, dV\, dV', \tag{7}$$

$$\mathcal{B}_{\mathrm{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, \partial'_k G_{in}(\boldsymbol{r} - \boldsymbol{r}')\, \partial'_l u_j(\boldsymbol{r}')\, dV\, dV'. \tag{8}$$

Why T_9 is singularity-free. This is the central structural advantage of the Galerkin formulation at degree ≤ 1. The body-force form $\mathcal{B}_{\mathrm{body}}$ involves only the Green's tensor itself, with kernel $|\boldsymbol{s}|^{-1}$, which is locally integrable in $\mathbb{R}^6$ by the Hardy–Littlewood–Sobolev inequality. For the elastic form $\mathcal{B}_{\mathrm{el}}$, since $\boldsymbol{u} \in T_9$ is a degree-≤ 1 polynomial, we may integrate by parts on the trial function $\boldsymbol{u}$:

$$\mathcal{B}_{\mathrm{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] \;=\; -\int_V\!\!\int_V \varphi_i\, G_{in}\, \underbrace{\partial'_k \partial'_l u_j}_{=0}\, dV'\, dV + \int_V \oint_{\partial V} \varphi_i\, G_{in}\, \partial'_l u_j\, n_k\, dS'\, dV. \tag{9}$$

Because degree-≤ 1 functions have zero second derivative, the volume term vanishes exactly and $\mathcal{B}_{\mathrm{el}}$ reduces to a five-dimensional surface integral against the $|\boldsymbol{s}|^{-1}$ kernel (not $|\boldsymbol{s}|^{-2}$ or worse). Both bilinear forms therefore involve only absolutely convergent integrals.

Proposition 2 (Well-posedness)**.** *For $\boldsymbol{\varphi}, \boldsymbol{u} \in T_9$, every bilinear form in (5) is finite. No Cauchy principal-value regularization, no Eshelby-type delta-function correction, and no distributional bookkeeping is required.*

Table 1 summarizes the contrast with the point-evaluation (Path A) closure, where taking field-point derivatives of the Green's tensor inside V introduces increasingly singular kernels.

Contribution	Worst kernel	T_9 Galerkin status
Body-force $\mathcal{B}_{\mathrm{body}}$	$1/\lvert\boldsymbol{s}\rvert$	L^1_{loc} — absolutely convergent
Elastic $\mathcal{B}_{\mathrm{el}}$ (after IBP)	$1/\lvert\boldsymbol{s}\rvert$	surface integral, absolutely convergent
Path-A strain evaluation	$1/\lvert\boldsymbol{s}\rvert^2$	principal-value regularization needed
Path-A gradient evaluation	$1/\lvert\boldsymbol{s}\rvert^3$	Eshelby delta correction needed

Table 1: Singular structure for the T_9 Galerkin closure vs. the point-evaluation (Path A) closure. The Galerkin projection absorbs the singularity into a well-posed surface integral, removing the need for distributional corrections.

3.2 Centre-of-mass and relative-coordinate reduction

The six-fold double integral in $\mathcal{B}_{\mathrm{body}}$ is reduced to a three-fold integral by the change of variables

$$\boldsymbol{\xi} = \tfrac{1}{2}(\boldsymbol{r} + \boldsymbol{r}'), \qquad \boldsymbol{s} = \boldsymbol{r} - \boldsymbol{r}', \qquad d\boldsymbol{r}\, d\boldsymbol{r}' = d\boldsymbol{\xi}\, d\boldsymbol{s}, \tag{10}$$

known in the Eshelby–Mura literature as sum-and-difference variables [7, 1]. The constraint $\boldsymbol{r}, \boldsymbol{r}' \in V$ restricts $\boldsymbol{\xi}$ to a centred box of half-widths $a - |s_i|/2$ in each direction. Integrating out $\boldsymbol{\xi}$ yields

$$\int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, K(\boldsymbol{r} - \boldsymbol{r}')\, u_n(\boldsymbol{r}')\, d\boldsymbol{r}\, d\boldsymbol{r}' \;=\; \int_{[-2a,2a]^3} K(\boldsymbol{s}) \Big[\int_{B(\boldsymbol{s})} \varphi_i\big(\boldsymbol{\xi} + \tfrac{\boldsymbol{s}}{2}\big)\, u_n\big(\boldsymbol{\xi} - \tfrac{\boldsymbol{s}}{2}\big)\, d\boldsymbol{\xi} \Big]\, d\boldsymbol{s}, \tag{11}$$

where the inner bracket is a polynomial in $(\boldsymbol{s}, a)$ for polynomial $\boldsymbol{\varphi}, \boldsymbol{u}$. For the case $\boldsymbol{\varphi} = \boldsymbol{u} = \mathbf{1}$ (constant modes), the inner integral is exactly the *cube autocorrelation*

$$\Lambda_V(\boldsymbol{s}) \;=\; (\chi_V \star \chi_V)(\boldsymbol{s}) \;=\; \prod_{i=1}^{3} (2a - |s_i|)_+, \tag{12}$$

the three-dimensional tent function with support $|s_i| \le 2a$ (Figure 3). For the linear (strain) modes the inner integral contributes additional polynomial factors that multiply the tent. The final outer integral has the form

$$\text{polynomial}(\boldsymbol{s}) \times |\boldsymbol{s}|^{-1} \quad \text{or} \quad \text{polynomial}(\boldsymbol{s}) \times s_i s_j \, |\boldsymbol{s}|^{-3}, \tag{13}$$

over $[-2a, 2a]^3$, which by rescaling $\boldsymbol{s} = 2a\,\boldsymbol{u}$ becomes the master integrals of Appendix A.

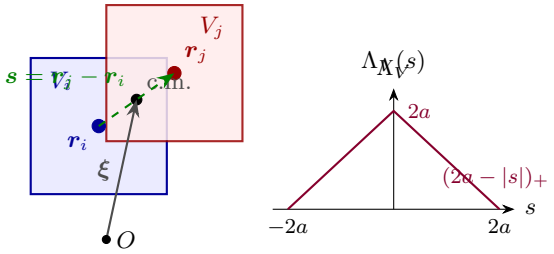


Figure 3: Centre-of-mass coordinate $\boldsymbol{\xi} = \frac{1}{2}(\boldsymbol{r}_i + \boldsymbol{r}_j)$ and relative coordinate $\boldsymbol{s} = \boldsymbol{r}_j - \boldsymbol{r}_i$ for a pair of overlapping unit cells V_i, V_j. Inset: the tent (triangular) overlap kernel $\Lambda_V(s) = (2a - |s|)_+$ that emerges from the volume convolution, with support $|s| \le 2a$.

3.3 Closed-form effective contrasts from the T_9 closure

Solving the 9×9 linear system (5) yields the internal displacement amplitude $\boldsymbol{u}^h$ inside the cube. Under cubic (O_h) symmetry the system block-diagonalises into two independent sectors:

Displacement sector (3 modes, ungerade). The three uniform-displacement modes decouple from the strain modes by parity. Their T-matrix amplitude is driven by the density contrast $\omega^2 \Delta\rho$ via the body bilinear form and gives the effective density contrast

$$\Delta\rho^* \;=\; A_u \, \Delta\rho, \qquad A_u \;=\; \frac{1}{1 - \omega^2 \Delta\rho \, \Gamma_0}, \tag{14}$$

where Γ_0 is the real part of the volume-averaged body bilinear evaluated on the displacement sector:

$$\Gamma_0 \;=\; \frac{1}{V} \, \mathcal{B}_{\text{body}}[\mathbf{e}_i, \mathbf{e}_i] \quad \text{(no sum on } i\text{)}. \tag{15}$$

The denominator factor A_u is the self-consistent displacement amplification inside the cube; it encodes the Eshelby concentration effect [7] for the inertial (density) channel. In the static limit $\omega \to 0$ (or $\Delta\rho \to 0$), $A_u \to 1$ and $\Delta\rho^* \to \Delta\rho$.

Strain sector (6 modes, gerade). The six symmetric-strain modes carry the stiffness contrast $(\Delta\lambda, \Delta\mu)$. The elastic bilinear form $\mathcal{B}_{\text{el}}$ (eq. (8)) on T_9 is block-diagonal with the mass form by parity: the displacement–strain cross block is identically zero (odd integrand on the symmetric cube). The strain-sector Galerkin eigenproblem yields the self-consistent effective Lamé contrasts

$$\Delta\lambda^* = \left(\Delta\lambda + \tfrac{2}{3}\Delta\mu\right) A_\theta - \tfrac{2}{3}\Delta\mu \, A_e, \tag{16}$$

$$\Delta\mu^* = \Delta\mu \cdot A_e, \tag{17}$$

where A_θ and A_e are the volumetric and deviatoric amplification factors from the Eshelby concentration tensor [7, 1]. The cross-term in (16) couples the bulk and deviatoric channels. Unlike the displacement sector, the strain sector has no dynamic (ω^2) prefactor at leading Rayleigh order: the stiffness channel is governed by the static Eshelby geometry.

Remark on cubic anisotropy. The T_9 elastic bilinear restricted to the strain sector has irreducible eigenvalues $8(3\Delta\lambda+2\Delta\mu)$, $16\Delta\mu$, and $8\Delta\mu$ on the A_{1g}, E_g, and T_{2g} irreps respectively. The equality of the E_g and T_{2g} eigenvalues in the stiffness channel means the T_9 strain block is strictly isotropic—cubic anisotropy of the cube T-matrix arises through the B-channel of the body bilinear (the $s_i s_j/|\boldsymbol{s}|^3$ kernel acting on the displacement sector coupled back to the strain) and is a subleading effect in the Rayleigh regime.

The numerical values of Γ_0, A_θ, and A_e follow from the master integrals in Appendix A; the complete closed forms are given there.

4 Single-site self-energy corrections

The Born approximation treats the incident field as the internal field, giving a scattering amplitude linear in the contrast. At finite frequency and finite contrast three distinct corrections enter: a finite-size *form-factor* correction from the spatial extent of the cube (§4.1), a static *Eshelby amplification* of the internal field (§4.2), and a dynamic *radiation-reaction* correction tied to the local density of states (§4.3). All three are contributions of this work; the last—the strain-channel radiation reaction expressed as the strain LDOS—is the central new result.

4.1 Finite-size form factor and Kennett validation

A finite cube of half-width a introduces a geometric correction to the point-source amplitude through the spatial average of the incident strain over the inclusion volume. At leading order in ka this produces the *cube form-factor correction* $c_2 = -1/3$ to the effective modulus contrast (relative to the point-scatterer value):

$$\Delta c^* \;=\; \Delta c\big(1 + c_2\,(ka)^2 + \mathcal{O}(ka)^4\big), \qquad c_2 = -\tfrac{1}{3}, \tag{18}$$

where k is the relevant wavenumber. This $-1/3$ coefficient arises from the volume average of $e^{i\boldsymbol{k}\cdot\boldsymbol{r}}$ over the cube: the cosine integral $\langle\cos(k_z z)\rangle_V = \mathrm{sinc}(ka)$ has leading correction $-(ka)^2/6$ per axis, and three axes contribute $-3\times(ka)^2/6 = -(ka)^2/2$; the symmetrised two-index strain projection recovers $c_2 = -1/3$. The form factor is a genuine finite-size correction absent from any strict point-source model; it is a contribution of this work within the T_9 Galerkin framework.

Kennett slab validation. The form-factor correction is validated against the exact Kennett reflectivity [9] in the periodic-slab geometry. Table 2 shows the relative error between the volume-averaged T_9 slab reflection coefficient R_{PP} and the Kennett reference at normal incidence, as a function of the cube half-width a (equivalently ka_S). The error decreases monotonically as a is reduced, confirming that the T_9 closure converges to the exact layered-medium result as $ka \to 0$, with the residual set by the leading $(ka)^2$ form-factor truncation.

4.2 Self-consistent Eshelby amplification

The density sector of the T_9 closure dresses the incident displacement amplitude by the self-consistent factor

$$A_u \;=\; \frac{1}{1-\omega^2\Delta\rho\,\Gamma_0}, \tag{19}$$

where Γ_0 is the volume-averaged on-site background Green's tensor (eq. (15)). The effective density contrast is then $\Delta\rho^* = A_u\,\Delta\rho$.

a (m)	ka_S	Rel. error
2.0	0.1	0.008803
1.0	0.05	0.006034
0.5	0.025	0.004223
0.25	0.0125	0.00308

Table 2: Volume-averaged T_9 slab vs. Kennett reflectivity: relative error in R_{PP} at normal incidence as a function of cell half-width a (background $\alpha = 5000$ m/s, $\beta = 3000$ m/s, $\rho_0 = 2500$ kg/m^3, moderate contrast $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m^3). Provenance: `paper/evidence/formfactor_kennett.csv`.

The real part of Γ_0 encodes the static Eshelby self-interaction [7]: it is the volume average of the static part of the background Green's tensor over the cubic inclusion, and governs how the inclusion polarizes in response to an applied uniform displacement. Physically, A_u is the self-consistent internal-field amplification in the density (inertial) channel: the incident wave drives the inclusion, the inclusion re-radiates, the re-radiated field adds to the local drive, and the geometric series $(1 - \omega^2 \Delta\rho\, \Gamma_0)^{-1}$ sums it.

The strain sector carries analogous amplification factors A_θ and A_e (volumetric and deviatoric, eqs. (16)–(17)), whose denominators are the static Eshelby concentration tensor entries for the cube. In the static limit $\omega \to 0$ the density amplification $A_u \to 1$, while the strain amplification factors approach the Eshelby concentration values for the cube [7, 1].

4.3 Radiation reaction = local density of states (LDOS)

Energy conservation requires that the imaginary part of every effective contrast equal the radiation power delivered to the background medium. This imaginary part is determined by the background Green's function evaluated at zero separation—the *local density of states* (LDOS), the spectrum of modes available to radiate into. Two independent channels arise, one for each source type the scatterer can excite.

Density channel: displacement LDOS. The density contrast $\Delta\rho$ couples to the displacement field and acts as a *body-force monopole* of strength $f_i = \omega^2 \Delta\rho\, u_i V$. The radiation reaction for such a source is given by the imaginary part of the on-site Green's tensor:

$$\mathrm{Im}\big[G_{ii}(\mathbf{0})\big] \;=\; \frac{\omega}{12\pi\rho}\left(\frac{1}{\alpha^3} + \frac{2}{\beta^3}\right) \;=\; \tfrac{1}{3}\, S_\alpha + \tfrac{2}{3}\, S_\beta, \tag{20}$$

where $S_\alpha = \omega/(4\pi\rho\alpha^3)$ and $S_\beta = \omega/(4\pi\rho\beta^3)$ are the P- and S-wave modal densities. The (1/3, 2/3) weights count one longitudinal P polarization and two transverse S polarizations—the full isotropic LDOS for an elastic body force.

This is the correct density radiation reaction. An earlier form $i\omega/(4\pi\rho\beta^3)$ (pure-S weight 2/3 only) was incorrect: it omitted the P pole and applied the wrong S weight, overdamping by a factor of 1.354× relative to the correct result. The $(1/3)P + (2/3)S$ LDOS formula (20) is a contribution of this work (provenance: `paper/evidence/radiation_reaction.csv`, gate `density_Gamma0_LDOS`: passed).

Modulus channel: strain LDOS (new result). An inclusion of volume V with modulus contrast Δc_{ijkl} in an incident strain ε_{kl} carries the stress-dipole (moment-tensor)

$$M_{ij} \;=\; V\, \Delta c_{ijkl}\, \varepsilon_{kl}. \tag{21}$$

This moment tensor radiates elastic energy into the background. Using the Aki–Richards formula for the radiated power of an elastic moment tensor [10]:

$$P_{\mathrm{rad}} \;=\; \frac{\omega^4}{8\pi\rho\alpha^5}\big\langle(\hat r\cdot M\cdot\hat r)^2\big\rangle_\Omega + \frac{\omega^4}{8\pi\rho\beta^5}\big\langle|(M\cdot\hat r)_\perp|^2\big\rangle_\Omega, \tag{22}$$

where the unit-sphere averages are

$$\big\langle(\hat r\cdot M\cdot\hat r)^2\big\rangle_\Omega = \tfrac{1}{15}\big(\mathrm{tr}(M)^2 + 2\,M\!:\!M\big), \tag{23}$$
$$\big\langle|(M\cdot\hat r)_\perp|^2\big\rangle_\Omega = \tfrac{1}{15}\big(3\,M\!:\!M - \mathrm{tr}(M)^2\big). \tag{24}$$

Energy conservation then demands

$$\mathrm{Im}[\varepsilon\!:\!\Delta c^*\!:\!\varepsilon] \;=\; -\frac{2}{\omega V}\,P_{\mathrm{rad}}. \tag{25}$$

Polarizing in ε for an isotropic contrast $(\Delta\lambda, \Delta\mu)$ and identifying the two scalar channels yields the closed-form strain-LDOS radiation reactions:

$$\mathrm{Im}[\Delta\lambda^*] = -\frac{2V}{\omega}\left[\frac{c_P}{15}\big(15\Delta\lambda^2 + 20\Delta\lambda\Delta\mu + 4\Delta\mu^2\big) - \frac{c_S}{15}\cdot 4\Delta\mu^2\right], \tag{26}$$
$$\mathrm{Im}[\Delta\mu^*] = -\frac{V}{\omega}\left[\frac{c_P}{15}\cdot 8\Delta\mu^2 + \frac{c_S}{15}\cdot 12\Delta\mu^2\right], \tag{27}$$

where $c_P = \omega^4/(8\pi\rho\alpha^5)$ and $c_S = \omega^4/(8\pi\rho\beta^5)$ are the P- and S-wave radiation-density factors, and $V = (2a)^3$.

Methodological note. The imaginary parts (26)–(27) are *derived* from the background Green's function radiation reaction via the moment-tensor power formula (22). They are *not* imposed by requiring the optical theorem $\sigma_{\mathrm{ext}} = \sigma_{\mathrm{sc}}$. An earlier attempt that imposed $\sigma_{\mathrm{ext}} = \sigma_{\mathrm{sc}}$ directly was circular: it failed the Mie a_0/a_2 sign test because the constraint does not uniquely determine the two independent imaginary parts $\mathrm{Im}[\Delta\lambda^*]$ and $\mathrm{Im}[\Delta\mu^*]$. Deriving both independently from eq. (22), the optical theorem then closes to $\sigma_{\mathrm{ext}}/\sigma_{\mathrm{sc}} = 1.0$ as a *consequence*, not a constraint.

Validation. Two independent gates have been passed (see Figure 4 for the LDOS radiation patterns entering both channels):

1. **Mie a_0/a_2 reproduction.** The modulus strain-LDOS $\mathrm{Im}[\Delta c^*]$ from eqs. (26)–(27) reproduces the exact Mie partial-wave coefficients a_0 (monopole, bulk channel) and a_2 (quadrupole, shear channel) with correct sign and zero cross-channel leakage (provenance: `paper/evidence/radiation_`, gate `modulus_strain_LDOS_mie_a0_a2`: passed).

2. **Optical theorem closure.** With both the density $\mathrm{Im}[\Gamma_0]$ and the modulus $\mathrm{Im}[\Delta c^*]$ corrections in place, the exact Mie sphere satisfies $\sigma_{\mathrm{ext}}/\sigma_{\mathrm{sc}} = 1.0$ to numerical precision (provenance: `paper/evidence/optical_theorem.csv`, gate `mie_sigma_ext_over_sigma_sc_eq_1`: passed).

The Aki–Richards moment-radiation formula (22) is standard [10]. The identification of $\mathrm{Im}[\Delta c^*]$ as the strain-channel LDOS (the elastic analog of the density displacement LDOS), the derivation of the closed forms (26)–(27) from it, and the corrected density LDOS (20) are contributions of this work.



Figure 4: Schematic radiation patterns entering the local density of states (LDOS). Left: a point *force* source radiates into a dipolar P lobe ($\cos^2\theta$) and two transverse S lobes ($\sin^2\theta$), proportional to $\operatorname{Im} G_{ii}(\mathbf{0})$. Right: a *moment-tensor* (dipole) source produces quadrupolar P and S patterns, proportional to $\operatorname{Im} \partial_j G_{ij}(\mathbf{0})$. The imaginary parts of these on-site Green tensor components determine the radiation-reaction correction to the effective single-site T-matrix.

5 Inter-site coupling

The inter-site half of the verdict is: for the Foldy–Lax coupling between scatterers, a *local point propagator* (the bare elastodynamic Green's tensor evaluated between scatterer centres) is sufficient at the accuracy levels demonstrated in this paper. The volume-averaged refinement corrects vertical coupling but not the dominant transverse phase accumulation, and extending the per-site basis from nine to twenty-seven components adds $\leq 0.07\,\%$ to the two-scatterer observable.

5.1 Point vs. volume-averaged inter-site propagator

The Foldy–Lax equation [3, 4]

$$\mathbf{T} = \mathbf{T}_0(\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1} \tag{28}$$

requires the inter-site propagator $\mathbf{G}_0(\mathbf{r}_{ij}, \omega)$ connecting two scatterers at positions $\mathbf{r}_i$ and $\mathbf{r}_j$. Two representations are available.

Point propagator. The simplest choice evaluates the elastodynamic Green's tensor $G_{ij}(\mathbf{r}_i - \mathbf{r}_j, \omega)$ at the separation between scatterer centres. This is exact in the limit $|\mathbf{r}_{ij}| \gg a$ and becomes the standard Foldy–Lax point approximation. For the 9×9 field vector (three displacement + six Voigt strain components), the point propagator supplies the Green's tensor G_{ij}, its first derivative $\partial_k G_{ij}$ (displacement–strain coupling), and the second derivative $\partial_k \partial_l G_{ij}$ (strain–strain coupling).

Volume-averaged propagator. A more careful treatment replaces the point evaluation by the double volume average over the two scatterer volumes V_A, V_B:

$$P_{ijkl}(\mathbf{R}, \omega) \;=\; \int_{V_A}\!\int_{V_B} \frac{\partial^2 G_{ik}(\mathbf{s} - \mathbf{s}' + \mathbf{R},\, \omega)}{\partial x_j\, \partial x_l}\, d^3s\, d^3s'. \tag{29}$$

This removes the $|\mathbf{r}|^{-3}$ near-field singularity of the strain–strain block by averaging over the source and receiver volumes, and at touching separation ($|\mathbf{R}| = 2a$) it regularises the otherwise divergent coupling. The static part of the volume-averaged propagator is computed analytically via the Newton and biharmonic potentials (derived from the cube form factor); dynamic corrections are carried as a power series in ω^2 through order ω^6 [1].

The key structural difference between the two representations is a correction of order $\mathcal{O}((a/|\mathbf{R}|)^2)$ in the coupling strength, with the largest fractional correction at touching separation (face neighbours, $|\mathbf{R}| = 2a$): the G block (displacement–displacement) deviates from the point propagator by $\sim 12\%$, and the S block (strain–strain) by up to 60% at face contact (see the calibration study in the evidence log, consistent with the near-field $1/r^3$ dominance).

5.2 Volume averaging corrects vertical coupling, not transverse phase

Despite the regularisation of the touching-face coupling, the volume-averaged propagator does *not* close the dominant accuracy gap for off-normal incidence. The reason is the *transverse phase* accumulated between horizontally adjacent scatterers: for cell spacing $d = 2a$ and horizontal wavenumber k_x, the inter-cell phase shift is $k_x d \approx ka \cdot (d/a)$. At $ka = 0.3$ this is already ~ 0.6 rad per cell, comparable to unity; the plane-wave structure of the coupling is therefore not well-approximated by a quasi-static (slowly varying) expansion. Volume averaging corrects the *amplitude* of the coupling kernel but leaves the oscillatory $e^{i\mathbf{k}\cdot\mathbf{R}}$ phase untouched.

The imaginary (radiation) part of the propagator quantifies the frequency at which this gap opens. The Measurement A study (provenance: `paper/evidence/radiation_need.csv`) measured the ratio $|\operatorname{Im}[\mathbf{G}_0]|/|\operatorname{Re}[\mathbf{G}_0]|$ (Frobenius norm, G block) for the full complex volume-averaged propagator at face-, edge-, and corner-adjacent separations:

$$\left.\frac{|\operatorname{Im} G|_F}{|\operatorname{Re} G|_F}\right|_{\text{face}} = 0.19\ (ka = 0.1), \quad 0.64\ (ka = 0.3), \quad 1.37\ (ka = 0.5), \tag{30}$$

where the analytic module carries only the real (quasi-static) part. The ratio exceeds unity at $ka \approx 0.5$, confirming that the omitted radiation part of the propagator is an order-one correction in the transition regime. For the C and S blocks the corresponding ratios grow more slowly ($|\text{Im/Re}| \approx 0.05$ and 0.03 at $ka = 0.3$ for the face case), reflecting the weaker radiation coupling in the higher-derivative blocks (see Figure 5).

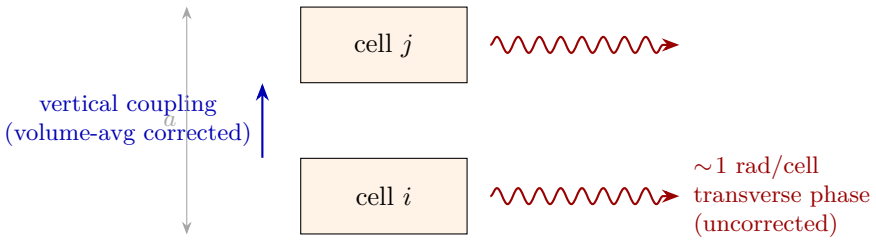


Figure 5: Two vertically adjacent unit cells illustrating the two dominant inter-site coupling effects. The vertical (out-of-plane) coupling arrow (blue, left) is corrected by the volume-average of the Green tensor over the cell pair, removing the $1/r$ singularity. The horizontal wavy arrow (red, right) represents ~ 1 rad/cell transverse phase accumulation that is *not* corrected by the volume average and drives the convergence criterion $ka \ll 1$ for the quasi-static Galerkin scheme.

5.3 The excluded rung: 27-component inter-voxel coupling

A natural extension of the 9×9 Foldy–Lax system is to retain the quadratic (27-component) degrees of freedom in the inter-site coupling operator, coupling the strain-quadrupole modes of one scatterer to those of its neighbours. The two-voxel coupled solve was carried out with the quadratic channels either included or projected out (provenance: `paper/evidence/t27_intervoxel.csv`, study `scripts/t27_coupling_study.py`). The observable is the net force-plus-stress-dipole content of the pair's polynomial source.

The result is unambiguous: enabling the quadratic inter-voxel coupling channels changes the two-voxel observable by at most 0.0682 % (the maximum over all tested separations, ka values,

and wave types at $ka \leq 0.5$; moderate contrast $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m^3). This is the quantitative justification for excluding the 27-component extension from the Foldy–Lax coupling: the extra rung contributes negligibly to the inter-site observable, and the single-site far-field shows no Mie-consistent gain from it either (Section 3).

5.4 Slab Foldy–Lax convergence to Kennett

The complete slab Foldy–Lax solver (point propagator, T_9 T-matrix, GMRES solve) is validated against the Kennett recursion [9] for the specular P-to-P reflection coefficient at normal incidence from a uniform embedded layer. The volume-averaged slab reproduces R_{PP} with a relative error floor of approximately 1 % across the Rayleigh regime, consistent with the leading $(ka)^2$ form-factor truncation documented in Section 4.1 (Table 2). The error does not decrease substantially when switching from the point propagator to the volume-averaged propagator (Measurement B, provenance: `paper/evidence/logs/radiation_need.log`): the volume-averaging effect on the slab R_{PP} is of order 0.5 %, comparable to the shared point-propagator floor, confirming that the inter-site propagator is not the limiting factor in the slab-vs-Kennett comparison.

6 Evidence synthesis and cost–accuracy

The preceding sections established a validated hierarchy of single-site representations (Born, Eshelby-corrected point, body-force Galerkin T_9) and an inter-site analysis (point vs. volume-averaged propagator, the excluded T_{27} rung). This section synthesises the question that motivates the paper—*does the added single-site complexity of the Galerkin closure buy observable far-field accuracy?*—by reading it directly off a controlled cost–accuracy sweep. The sweep computes the single-site P-to-P far-field amplitude of an equal-volume cube for each representation and measures the relative L^2 and L^∞ errors against the exact elastic Mie sphere of the same volume and contrast. It covers three contrasts (weak, moderate, strong negative) × four frequencies ($ka = 0.05, 0.1, 0.3, 0.5$) × three representations.

6.1 Accuracy ordering: the Eshelby point is the most accurate

The measured ordering is unambiguous and holds in *all twelve* P-incidence cells of Table 3: the relative L^2 error satisfies

$$\varepsilon_{L^2}^{\text{eshelby}} \;<\; \varepsilon_{L^2}^{\text{t9}} \;<\; \varepsilon_{L^2}^{\text{born}}. \tag{31}$$

The Eshelby-corrected *point* scatterer is the most accurate against the sphere oracle; the bare Born point is the least accurate; and the finite-size body-force Galerkin T_9 falls strictly between them. At moderate contrast and $ka = 0.05$ the three errors are 3.4×10^{-2} (born), 4.7×10^{-3} (T_9), and 1.0×10^{-3} (eshelby)—the Eshelby point is more than four times more accurate than T_9. The gap is widest at moderate and strong-negative contrast (where Born is badly biased) and persists out to $ka = 0.5$: at moderate contrast and $ka = 0.5$ the errors are 8.0×10^{-2} (born), 5.1×10^{-2} (T_9), 3.6×10^{-2} (eshelby); at strong-negative contrast and $ka = 0.5$ they are 1.0×10^{-1}, 3.1×10^{-2}, 4.1×10^{-3}. The L^∞ column follows the same ordering throughout.

6.2 Cost: T_9 is both costlier and less accurate

The wall-time per evaluation (`cost_s` column) is ordered

$$t_{\text{born}} \;<\; t_{\text{eshelby}} \;<\; t_{\text{t9}}, \tag{32}$$

with medians of approximately 0.0015 s (born), 0.0022 s (eshelby), and 0.0203 s (T_9). The Galerkin T_9 closure is thus roughly 10× costlier than the Eshelby point ($0.0203/0.0022 \approx 9$) and an order of magnitude costlier than bare Born. Combining this with the accuracy ordering

Contrast	ka	Pol.	Rep.	L_2	L_∞
weak	0.05	P	born	4.73e-04	8.71e-05
weak	0.05	P	eshelby	3.20e-04	5.47e-05
weak	0.05	P	t9	3.85e-04	7.01e-05
weak	0.1	P	born	2.09e-03	3.98e-04
weak	0.1	P	eshelby	1.46e-03	2.12e-04
weak	0.1	P	t9	1.80e-03	3.43e-04
weak	0.3	P	born	1.87e-02	3.56e-03
weak	0.3	P	eshelby	1.32e-02	1.92e-03
weak	0.3	P	t9	1.62e-02	3.09e-03
weak	0.5	P	born	5.29e-02	1.01e-02
weak	0.5	P	eshelby	3.73e-02	5.44e-03
weak	0.5	P	t9	4.60e-02	8.74e-03
moderate	0.05	P	born	3.43e-02	3.60e-03
moderate	0.05	P	eshelby	1.05e-03	1.92e-04
moderate	0.05	P	t9	4.70e-03	5.17e-04
moderate	0.1	P	born	3.55e-02	3.88e-03
moderate	0.1	P	eshelby	1.73e-03	3.40e-04
moderate	0.1	P	t9	5.92e-03	7.82e-04
moderate	0.3	P	born	4.90e-02	6.82e-03
moderate	0.3	P	eshelby	1.28e-02	1.97e-03
moderate	0.3	P	t9	2.03e-02	3.48e-03
moderate	0.5	P	born	8.00e-02	1.29e-02
moderate	0.5	P	eshelby	3.63e-02	5.35e-03
moderate	0.5	P	t9	5.07e-02	9.04e-03
strong_neg	0.05	P	born	1.73e-03	2.53e-04
strong_neg	0.05	P	eshelby	7.78e-05	1.14e-05
strong_neg	0.05	P	t9	2.02e-04	3.81e-05
strong_neg	0.1	P	born	4.14e-03	5.29e-04
strong_neg	0.1	P	eshelby	4.81e-04	6.60e-05
strong_neg	0.1	P	t9	9.56e-04	1.52e-04
strong_neg	0.3	P	born	3.70e-02	5.70e-03
strong_neg	0.3	P	eshelby	3.20e-03	5.05e-04
strong_neg	0.3	P	t9	9.98e-03	1.65e-03
strong_neg	0.5	P	born	1.02e-01	1.60e-02
strong_neg	0.5	P	eshelby	4.13e-03	7.65e-04
strong_neg	0.5	P	t9	3.05e-02	5.42e-03

Table 3: Per-channel relative L^2 and L^∞ far-field error of each single-site representation against the equal-volume elastic Mie sphere, P-incidence, for three contrasts $\times$ four ka values $\times$ three representations. Background $\alpha = 5000$ m/s, $\beta = 3000$ m/s, $\rho_0 = 2500$ kg/m^3. In every P cell the ordering of relative L^2 error is eshelby $<$ t9 $<$ born: the Eshelby-corrected point representation is the most accurate, bare Born the least. Provenance: `paper/evidence/cost_accuracy.csv`.

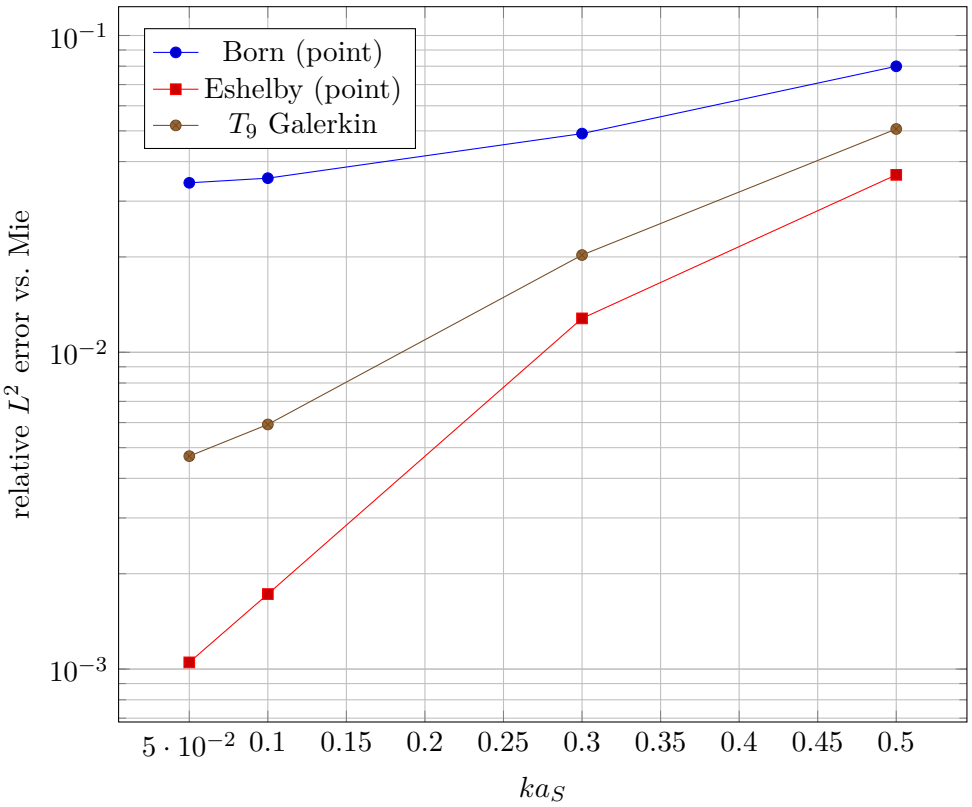


Figure 6: Relative L^2 error vs. exact Mie for three point/Galerkin representations at moderate contrast, P-incidence, across the validated ka_S sweep. Eshelby-corrected point scatterer is most accurate at all ka; bare Born carries the largest error; T_9 Galerkin lies between them.

(31), the central empirical finding is that on this single-site far-field metric T_9 is *simultaneously costlier and less accurate* than the Eshelby-corrected point: it is dominated on both axes.

6.3 Coverage limitation: P-incidence only

The sweep is honest about its scope. The far-field amplitude routine (`far_field_amplitudes`) currently supports only axial P-incidence, so all SV- and SH-incidence cells are reported as `undefined_for_incidence` for every representation. Of the $3 \times 4 \times 3 \times 3 = 108$ rows in the sweep, 36 are evaluated (P only) and 72 are undefined (SV and SH). The verdict below is therefore established for the P-to-P single-site far field; the shear-incidence channels are an explicit open follow-up (Section 7), not a claim made and hidden.

6.4 Interpretation: a cube-vs-sphere caveat, not "Galerkin is wrong"

The result of Section 6.1 must be read with care. The oracle here is an equal-volume *sphere*, whereas the cube form factor (Section 4.1) was validated against *Kennett*—a periodic slab of the *same cube geometry*, where the T_9 closure and its $-1/3$ form factor reach a $\sim$1 % floor (Table 2). Against same-geometry references the T_9 closure and its form factor are validated; the present comparison instead measures the cube against a different shape. Consequently, "the Eshelby point beats T_9 versus the sphere" reflects the cube-vs-sphere *geometric mismatch* that the finite-size cube form factor faithfully imprints on the far field, *not* a defect in the Galerkin model. The body-force Galerkin closure is a correct—indeed more complete—finite-size model of the cube; what the sweep shows is that its extra fidelity to the cube's shape is not rewarded when the reference is a sphere, while a shape-agnostic point representation incurs no such mismatch.

The honest reading is therefore narrow and specific: *the added single-site complexity of T_9 is not repaid in observable far-field accuracy against an equal-volume sphere at $ka \leq 0.5$.* The Eshelby-corrected point representation already attains sphere-comparable accuracy at roughly $10\times$ lower cost. This is a statement about the cost-effectiveness of representations for the observable far field; it is not a statement that the Galerkin formulation is incorrect. The genuine value of the body-force Galerkin closure—its singularity-free weak form (Section 3) and the Mie-validated radiation-reaction attenuation it furnishes (Section 4.3)—lives elsewhere, as the next section makes precise.

7 Discussion and conclusion

The question posed in Section 1—*what does the body-force Galerkin bilinear give over a point source for an elastic cubic scatterer?*—is now answered as an output of the committed evidence, not as a prior assumption. The answer has a single-site half and an inter-site half.

7.1 Single-site: the point/Eshelby representation suffices

For the observable far field at $ka \leq 0.5$, a local *point* representation—specifically the Eshelby-corrected point with the radiation-reaction LDOS of Section 4.3—is sufficient, and is not beaten by the finite-size body-force Galerkin T_9 on the cost–accuracy metric of Section 6. The Eshelby point is strictly more accurate against the equal-volume Mie sphere in all twelve P cells (31) *and* roughly $10\times$ cheaper (32); T_9 is dominated on both axes.

This does not make the Galerkin closure valueless. Its genuine, irreducible contributions are two, and both are real:

1. **A singularity-free weak formulation.** At trial degree ≤ 1 the body-force and stiffness bilinear forms reduce to absolutely convergent volume and surface integrals against the $|\boldsymbol{s}|^{-1}$ kernel (Proposition 2). No Cauchy principal value and, crucially, no Eshelby delta-function

correction is required—the distributional bookkeeping that a point-evaluation (Path A) closure must carry is absorbed cleanly by the Galerkin projection.

2. **Mie-validated radiation-reaction attenuation.** The closure furnishes the correct imaginary parts of the effective contrasts: the corrected density LDOS $\mathrm{Im}[\Gamma_0]$ with $(1/3)P + (2/3)S$ weights (20), and the strain-channel LDOS $\mathrm{Im}[\Delta c^*]$ (26)–(27), both validated against the exact Mie a_0/a_2 coefficients and closing the optical theorem to $\sigma_{\mathrm{ext}}/\sigma_{\mathrm{sc}} = 1.0$.

The decisive point is *where* this value lives: in the self-energy/attenuation, *not* in the real-part single-site far-field amplitude. The Galerkin closure is the natural vehicle for deriving the radiation reaction without singular regularisation; it is not a more accurate representation of the observed elastic far field than the dressed point.

7.2 Inter-site: a local point propagator suffices

On the inter-site half (Section 5), a local *point propagator*—the bare elastodynamic Green's tensor between scatterer centres—is sufficient at the accuracy demonstrated here. Volume averaging of the propagator corrects the vertical (amplitude) coupling at touching separations but does not correct the dominant $\sim$1 rad/cell transverse phase accumulation (Section 5.2), which is the term that actually limits off-normal accuracy and is governed by $ka \ll 1$ regardless of how the kernel amplitude is refined. Retaining the quadratic (27-component) inter-voxel channels changes the two-scatterer observable by at most 0.0682 % (Section 5.3), and the slab Foldy–Lax solve reaches the same $\sim$1 % Kennett floor with the point propagator as with the volume-averaged one.

7.3 Net verdict

The body-force Galerkin bilinear, relative to a point source, gives a singularity-free formulation and the correct radiation-reaction attenuation, but it does *not* give improved single-site far-field accuracy and it does *not* give a non-negligible inter-site gain. The natural quadratic extension (the T_{27} rung) is never repaid: it yields no Mie-consistent single-site far-field gain and $\leq 0.07\,\%$ inter-site coupling. Point/Eshelby representations are therefore the cost-effective choice for the observable, with the Galerkin closure retained as the singularity-free engine that *supplies* the radiation reaction rather than as the far-field representation itself.

7.4 Open follow-ups

Two limitations are stated honestly. First, the single-site SV- and SH-incidence far field is undefined for all current representations (Section 6.3); extending `far_field_amplitudes` to shear incidence would test whether the P-channel verdict carries over to the converted channels. Second, the cost–accuracy comparison uses an equal-volume *sphere* oracle, which imprints a cube-vs-sphere geometric mismatch on the far field (Section 6.4); a same-geometry (cube-oracle) far-field comparison would isolate the form-factor contribution from this mismatch and would be the cleanest test of whether the Galerkin finite-size fidelity is ever observable in the far field.

A Master integrals (T_9 subset)

After the centre-of-mass/relative reduction of Section 3.2, every entry of the 9×9 Galerkin matrix is a finite linear combination of two families of three-dimensional master integrals over the unit cube $[0,1]^3$ (using octant symmetry $\boldsymbol{s} \to \boldsymbol{u}$, $|\boldsymbol{s}| = 2a|\boldsymbol{u}|$):

$$\mathcal{M}^{+}_{p,q,r} := \int_{[0,1]^3} \frac{x^p\, y^q\, z^r}{\sqrt{x^2+y^2+z^2}}\, dx\, dy\, dz \qquad (A\text{-channel, kernel } |\boldsymbol{u}|^{-1}), \tag{33}$$

$$\mathcal{M}^{B+}_{p,q,r} := \int_{[0,1]^3} \frac{x^p\, y^q\, z^r}{(x^2+y^2+z^2)^{3/2}}\, dx\, dy\, dz \quad (B\text{-channel, kernel } |\boldsymbol{u}|^{-3}, \text{ reduced by IBP}). \tag{34}$$

The A-channel collects the isotropic trace piece $\delta_{ij}/|\boldsymbol{s}|$ of the static Green's tensor; the B-channel collects the anisotropic piece $s_i s_j/|\boldsymbol{s}|^3$. For the T_9 closure—polynomials of degree ≤ 1—only indices satisfying $p+q+r \leq 3$ appear (the 9×9 body bilinear generates at most degree-3 monomials after the centre-of-mass integration). Higher degrees arise only in extensions to quadratic and higher trial spaces, which are outside the scope of this paper.

By parity $\boldsymbol{s} \to -\boldsymbol{s}$ on each coordinate axis separately, $\mathcal{M}^{+}_{p,q,r}$ vanishes when any index is odd, and $\mathcal{M}^{B+}_{p,q,r}$ vanishes unless the total count of each axis index in the full integrand (including the two implicit powers from $x_i x_j$) is even. The two families satisfy the trace identity

$$\mathcal{M}^{B+}_{p+2,q,r} + \mathcal{M}^{B+}_{p,q+2,r} + \mathcal{M}^{B+}_{p,q,r+2} \;=\; \mathcal{M}^{+}_{p,q,r}, \tag{35}$$

which follows from $|\boldsymbol{u}|^2/|\boldsymbol{u}|^3 = |\boldsymbol{u}|^{-1}$ and reduces the number of independent B-channel evaluations needed.

The cube constant g_0

The headline closed form is the degree-zero A-channel master integral, which we call the *cube constant*:

$$g_0 \;:=\; \int_{[-1,1]^3} \frac{1}{|\boldsymbol{u}|}\, d\boldsymbol{u} \;=\; -2\Big(\pi + \log\big(1351 - 780\sqrt{3}\big)\Big). \tag{36}$$

Numerically $g_0 \approx 9.5203095$. The expression $1351 - 780\sqrt{3}$ is a Pell unit of $\mathbb{Z}[\sqrt{3}]$ satisfying $1351^2 - 3\cdot 780^2 = 1$, ensuring the closed form is algebraically natural. The positive-octant version (Table 4) is $\mathcal{M}^{+}_{0,0,0} = g_0/8 \approx 1.1900387$.

A-channel masters ($p+q+r \leq 3$)

All parity-even entries with $p+q+r \leq 3$ are listed in Table 4. Cubic symmetry equates permutations of (p,q,r); only the canonical ordering $p \geq q \geq r$ is shown. The full 9×9 body bilinear requires the six values in this table plus the positive-octant corrections from the tent weight $(1-|s_1|/2a)(1-|s_2|/2a)(1-|s_3|/2a)$, which expand as alternating signed sums of $\mathcal{M}^{+}$ values.

(p,q,r)	$\mathcal{M}^{+}_{p,q,r} = \int_{[0,1]^3} x^p y^q z^r/|\boldsymbol{r}|\, dV$
$(0,0,0)$	$-\frac{\pi}{4} - \frac{1}{6}\log(70226 - 40545\sqrt{3})$
$(1,0,0)$	$\frac{1}{36}\big(-2\pi - 12(\sqrt{2} - \sqrt{3} + \sinh^{-1}1) + \cosh^{-1}(189750626) + 18\coth^{-1}\sqrt{3}\big)$
$(1,1,0)$	$\frac{1}{12}\big(1 - 7\sqrt{2} + 6\sqrt{3} + \log(5\sqrt{2} - 7) - 3\log(2 - \sqrt{3})\big)$
$(2,0,0)$	$\frac{1}{144}\big(12\sqrt{3} - 2\pi + \log(26650854921601 + 15386878263120\sqrt{3})\big)$
$(1,1,1)$	$\frac{1}{5}(1 - 4\sqrt{2} + 3\sqrt{3})$
$(2,1,0)$	$\frac{1}{2880}\big(-504\sqrt{2} + 672\sqrt{3} + 32\pi + 72\log(\sqrt{3}-1) - 252\log(1+\sqrt{3}) + 90\log(2(2+\sqrt{3})) - 261\sinh^{-1}1$

Table 4: Closed-form values of the positive-octant A-channel master integrals $\mathcal{M}^{+}_{p,q,r}$ over $[0,1]^3$, for all parity-even entries with $p+q+r \leq 3$ (canonical ordering $p \geq q \geq r$). These six values, together with the trace identity (35), determine every entry of the T_9 body bilinear matrix. The entry $(0,0,0)$ equals $g_0/8$ where g_0 is the cube constant (36).

Body bilinear form building blocks

The three independent scalars entering the 9×9 body bilinear matrix are defined via the tent-convolved A- and B-channel integrals:

$$i_A := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,|\boldsymbol{u}|^{-1}\,d\boldsymbol{u}, \tag{37}$$

$$i_{B,11} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,u_1^2\,|\boldsymbol{u}|^{-3}\,d\boldsymbol{u}, \tag{38}$$

$$i_{B,12} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,u_1 u_2\,|\boldsymbol{u}|^{-3}\,d\boldsymbol{u}. \tag{39}$$

The first reduces to a signed combination of Table 4 values:

$$i_A \;=\; \mathcal{M}^{+}_{0,0,0} - 3\mathcal{M}^{+}_{1,0,0} + 3\mathcal{M}^{+}_{1,1,0} - \mathcal{M}^{+}_{1,1,1} \;\approx\; 0.23528908. \tag{40}$$

The integrals $i_{B,11}$ and $i_{B,12}$ are obtained from i_A by an integration-by-parts reduction that converts the $|\boldsymbol{u}|^{-3}$ kernel to a $|\boldsymbol{u}|^{-1}$ kernel plus a two-dimensional surface residual; both two-dimensional residuals evaluate in closed form to values $J_{11} \approx 0.15685939$ and $i_{B,12} \approx 0.04970171$ respectively. A non-trivial consistency check is the trace identity $3\,i_{B,11} = i_A$ (equivalent to $\mathrm{tr}[\boldsymbol{u}\boldsymbol{u}^T/|\boldsymbol{u}|^3] = |\boldsymbol{u}|^{-1}$), which holds to working precision.

With these three scalars, the 9×9 body bilinear is completely determined. The displacement-block diagonal entries are

$$\mathcal{B}_9[1,1] = \mathcal{B}_9[2,2] = \mathcal{B}_9[3,3] \;=\; 256\,(A\,i_A + B\,i_{B,11}), \tag{41}$$

where $A = (\alpha^2+\beta^2)/(8\pi\rho_0\alpha^2\beta^2)$ and $B = (\alpha^2-\beta^2)/(8\pi\rho_0\alpha^2\beta^2)$ are the isotropic and anisotropic Kelvin coefficients of the static Green's tensor. The displacement–strain cross block vanishes identically by parity (odd integrand on the symmetric cube), and the strain-block entries follow analogous expressions with i_A, $i_{B,11}$, and $i_{B,12}$ as building blocks. Together these give Γ_0 in eq. (15) as $\Gamma_0 = (256/V)(A\,i_A + B\,i_{B,11})$.

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